

Xinxian Ma

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Education

Ecole Polytechnique Fédérale de Lausanne (EPFL), MS in Data Science Sept 2025 – Present

- **Course:** Machine Learning, Modern nature language processing, Applied Data Analysis

The Chinese University of Hong Kong (Shenzhen), BS in Data Science Sept 2021 – May 2025

- **GPA:** 3.9/4.0 (Top 2%)
- **Course:** Machine Learning, Deep Learning, Advanced Machine Learning

Research Experience

zip2zip-Based Compression-Aware Language Modeling Feb 2026 – Present

Research Assistant, Data Science & AI Lab (DLAB), EPFL

Lausanne, Switzerland

Supervisor: Prof. Robert West

- Develop and train compression-aware LLMs with inference-time adaptive tokenization, enabling 15–40% fewer input/output tokens while preserving competitive language modeling and downstream performance.
- Diagnose failure modes of compressed-token modeling and develop architectural and training improvements, including decoupled input/output representations, improved initialization, and span-aware RoPE positioning.
- Build efficient generation and evaluation infrastructure, including tree-attention-based batched decoding and multi-view evaluation, and speculative decoding; support end-to-end pretraining/SFT, LoRA and full fine-tuning on LLaMA- and Phi-based models.

Federated Data Visualization and Clustering

Apr 2024 – Aug 2024

Research Assistant, Shenzhen Research Institute of Big Data

Shenzhen, China

Supervisor: Prof. Jicong Fan

- Proposed Fed-tSNE and Fed-UMAP for privacy-preserving high-dimensional data visualization in federated learning, leveraging kernel methods and maximum mean discrepancy.
- Extended the framework to distributed clustering algorithms, including spectral clustering, and conducted large-scale experiments to evaluate effectiveness and scalability.

Publications

Federated t-SNE and UMAP for Distributed Data Visualization (**AAAI 2025**)

Sept 2024

Dong Qiao, *Xinxian Ma*, Jicong Fan

arXiv:2412.13495

Internship Experience

Data Scientist Intern, Flywheel – Shenzhen, China

Jun 2025 – Aug 2025

- Built standardized sales and media performance analytics pipelines using Amazon Marketing Cloud (AMC) SQL, enabling audience package creation and multi-dimensional cross-campaign benchmarking
- Produced custom insights and activation strategies for key brands by analyzing incremental customer acquisition, lifecycle value, and purchase paths to optimize targeting and media spend.

Data Scientist Intern, Anker – Shenzhen, China

Feb 2025 – Aug 2025

- Performed end-to-end EDA on sales data, including anomaly detection and treatment to improve data reliability for downstream modeling.
- Built demand forecasting pipelines using time-series models and LightGBM to support inventory planning.
- Developed product profile via clustering using wavelet denoising and DTW-based time-series K-means

AI Engineer Intern, Fandao – Guangzhou, China

Jun 2024 – Aug 2024

- Contributed to customized AI content generation by participating in fine-tuning, model testing, dataset preparation, and facilitated workflow design to ensure robust business solution implementation

- Assisted in development of an AI customer service system using Retrieval-Augmented Generation (RAG), enhancing product sales and department satisfaction

Projects

RPNR: Indirect Prompt Injection Benchmark for GitHub MCP Agents github.com/repo

- Built a multi-surface benchmark for indirect prompt injection attacks against GitHub MCP coding agents, covering repository files, issues, pull requests, and comments.
- Implemented an end-to-end evaluation harness with LangChain, LangGraph, OpenAI-compatible APIs, and the official GitHub MCP server, supporting tool whitelists, trajectory logging, retry, and recovery.
- Evaluated four LLM coding agents and two defenses on 384 tasks and 320 GitHub repositories, showing clean-task success above 90% while attack success reached up to 14.9%.
- Tools Used: Python, LangChain, LangGraph, GitHub MCP, LLM-as-Judge, JSON schemas.

Smiley Prediction in Twitter Sentiment Data github.com/repo

- Built a tweet sentiment classification pipeline using GloVe/FastText baselines and fine-tuned TweetNLP and BERTweet models.
- Achieved 92.0% on the public leaderboard (Rank #3).
- Tools Used: Python, PyTorch, Hugging Face Transformers, scikit-learn.

N-square queens problem with MCMC github.com/repo

- Implements Markov Chain Monte Carlo methods for the 3D (N^2)-Queens problem using the Metropolis–Hastings algorithm with simulated annealing, resulting in a stable and convergent solution approach.
- Tools Used: Python

Skill

Technical: Python, C++ , C, Java, R, Matlab, SQL, JavaScript

Language: English (IELTS 7.5), Mandarin (Native)

Award

Mathematical Contest in Modeling (Top 5%) Feb 2024

Dean’s List (2021, 2022, 2023, 2024) 2021 – 2024

CUHK(sz) Academic Scholarship 2021 - 2024

Others

Teaching Assistant, EPFL CS-401 Applied Data Analysis 2026 Fall

Teaching Assistant, CUHK(sz) STA2003 Statistics & Probability 2023 Fall

Aid Education Volunteer Certificate Of Appreciation soapcycling.org